

Northeastern University

**Principles and Applications of Remote
Sensing (Bilingual) Computer Practice
Report**



Professional: surveying and mapping

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1. Computer experiment content (1): band conversion, radiation correction

1. Experimental content

① Data preparation: Download Landsat or Sentinel images with an interval of more than five years between two scenes, and the image area is your hometown or other areas of interest. Requirements: At least one of the downloaded data is new data in recent years; Clear and good-quality images without clouds are needed; You need to try to download images in the same month or quarter; It needs to include 6 types of land use types: cultivated land, forest land, grassland, water body, construction land and unused land. If there are less than 6 types of land features in the study area, please select another area!

② Radiation calibration: Find the calibration parameters, select the radiation calibration tool, set the calibration parameters, and execute the radiation calibration

③ Atmospheric correction: Open the FLAASH module, input images and parameters, set output parameters, perform atmospheric correction, remove the influence of the atmosphere, and calculate the true reflectivity of the surface.

④ View the correction results: select at least three ground features and describe the differences in pixel values of the three ground features in different wavelength bands.

2. Experimental purpose

① Understand the band characteristics of remote sensing images: master the band composition and physical significance of multispectral remote sensing images, and understand the differences in the response of different bands to ground features.

② Master the principles and methods of radiation correction: learn the principles of radiation calibration and atmospheric correction, and master the technical process of eliminating sensor errors and atmospheric interference.

③ Proficient in remote sensing software operation skills: Complete band combination transformation and radiation correction operations through tools such as ENVI or ArcGIS to improve remote sensing data processing capabilities.

④ Cultivate quantitative analysis ability: analyze the influence of radiation correction on remote sensing quantitative inversion (such as vegetation index and surface temperature) by comparing images before and after correction.

3. Experimental principle

Band conversion principle

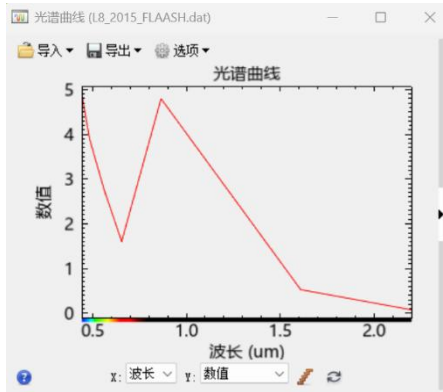
- 1. Band composition of multispectral remote sensing images:** Multispectral remote sensing images contain multiple bands with different wavelength ranges, and each band records the reflection or emission characteristics of ground objects in a specific spectral range. There are differences in reflectivity or radiance of different features in different wavelength bands, which is the basis of extracting feature information through wavelength band transformation. For example, vegetation has a high reflectivity in the near-infrared band, while water bodies have a low reflectivity in the near-infrared band and a relatively high reflectivity in the blue light band.
- 2. Band combination transformation:** By combining different bands, specific feature types can be highlighted or the contrast between features can be enhanced. Commonly used band combination methods include RGB false color synthesis, etc. For example, if the near-infrared, red light and green light bands are selected as RGB channels for synthesis respectively, the vegetation will appear bright red, the water body will be black or dark blue, and the construction land will appear off-white or blue-gray, which can be more intuitively identified. Different feature types are easy to interpret and analyze.

Radiation correction principle

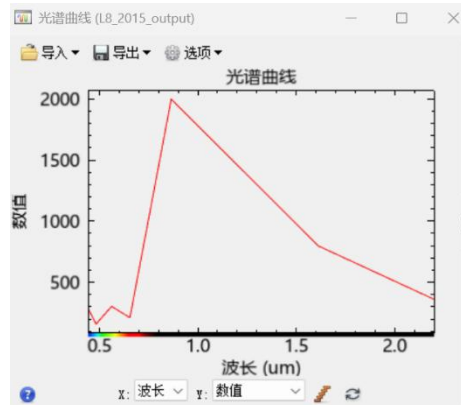
- 1. Radiation calibration principle:** When a sensor acquires remote sensing images, there is a certain relationship between its measured value and the actual radiation brightness of ground objects. Radiation calibration is the process of determining this quantitative relationship. By finding the calibration parameters of the sensor, the digital quantization value (DN value) of the sensor is converted into the absolute radiance value by using a specific formula. This can eliminate the error and inconsistency of the sensor itself, and make the data obtained by different sensors at different times comparable in terms of radiation amount. For example, for the sensors of Landsat series satellites, the calibration parameters will take into account factors such as the gain, offset and spectral response function of the sensor to accurately convert the DN value into radiance.
- 2. Atmospheric correction principle:** The atmosphere has a significant impact on the transmission of remote sensing signals, which will lead to light scattering and absorption, so that the radiation signals received by the sensor do not completely come from the real reflection on the surface of ground objects. The purpose of atmospheric correction is to eliminate the influence of the atmosphere and restore the true reflectivity of ground objects. The FLAASH module is a commonly used atmospheric correction method, which is based on radiative transfer theory and considers processes such as atmospheric molecular scattering, aerosol scattering and absorption. By inputting the relevant parameters of the image (such as sensor type, band information, imaging time, etc.) and atmospheric parameters (such as atmospheric mode, aerosol type and concentration, etc.), the transmission process

of radiation by the atmosphere is simulated, and the true reflectivity of the surface is obtained by inversion. In this way, the quality of remote sensing data can be improved, and quantitative analysis such as ground feature classification, vegetation index calculation and surface temperature inversion can be carried out more accurately.

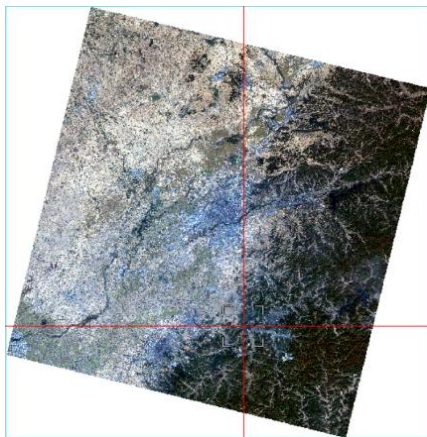
3、Experimental results and analysis



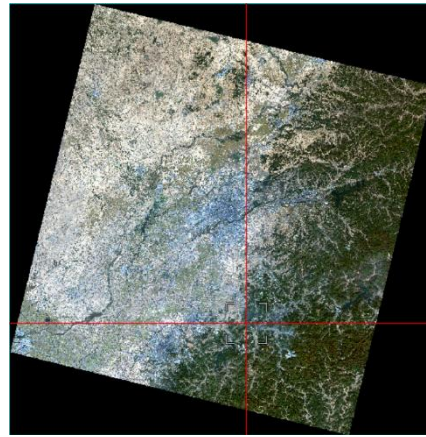
(a)Vegetation band curve before atmospheric correction



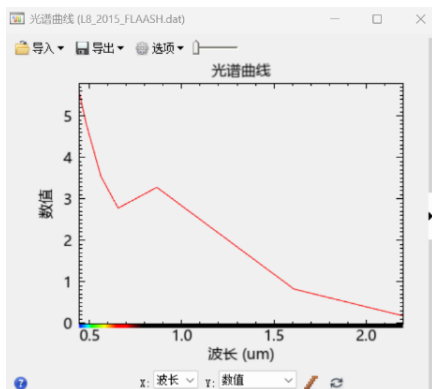
(b) Vegetation band curve after atmospheric correction



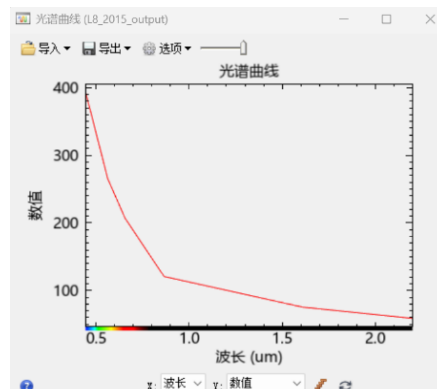
(a)Remote sensing image before after atmospheric correction



(b) Remote sensing image after atmospheric correction



(a)Water band curve before atmospheric correction



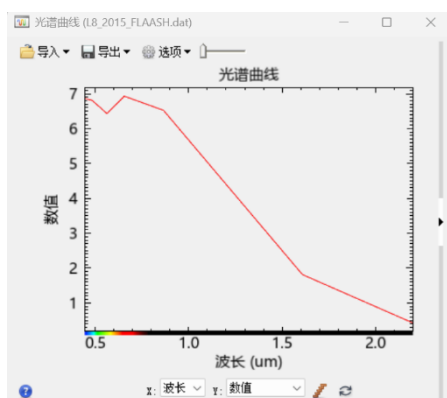
(b) Water band curve after atmospheric correction



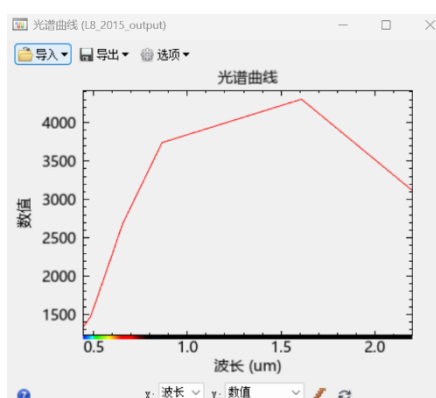
(a) Remote sensing image before atmospheric correction



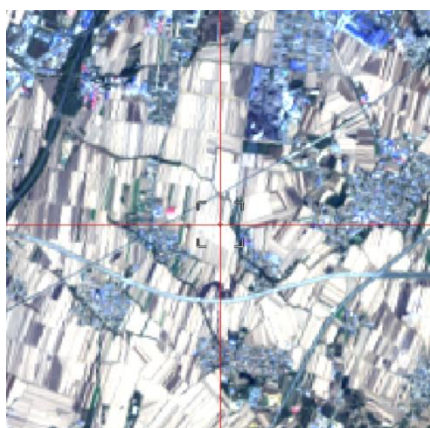
(b) Remote sensing image after atmospheric correction



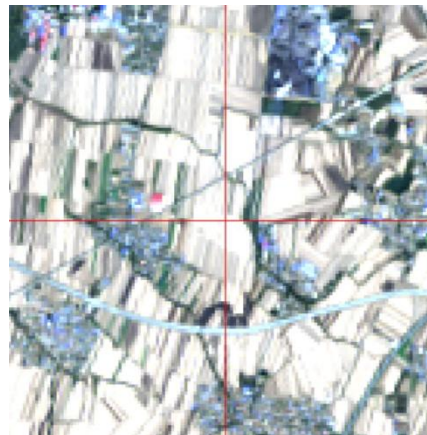
(a) Cultivated land band curve before atmospheric correction



(b) Cultivated land band curve after atmospheric correction



(a) Remote sensing image before atmospheric correction



(b) Remote sensing image after atmospheric correction

Experimental analysis:

1. Analysis of band conversion effect: In multispectral remote sensing image processing, band conversion is a key step. Through the combination of different wavelength bands, specific feature types can be highlighted or the contrast between features can be enhanced. Taking RGB false color synthesis (near-infrared, red light and green light bands are used as RGB channels respectively) as an example, vegetation appears bright red in this synthetic image. This is because vegetation has high reflectivity in the near-infrared band, moderate reflectivity in the red light band, and low reflectivity in the green light band. This combination distinguishes vegetation from other ground features in the image and helps to quickly identify and analyze vegetation distribution. The reflectivity of the water body is very low in the near-infrared band and relatively high in the blue light band. It is black or dark blue in the false color synthetic image, which can be clearly distinguished from other ground features and is convenient to define the scope of the water body. The construction land is gray-white or blue-gray, and it can also be clearly distinguished from the surrounding features, which is convenient for the study of urban expansion and land use change.

2. Analysis of radiation calibration effect: Radiation calibration converts the digital quantization value (DN value) of the sensor into the absolute radiance value, effectively eliminating the error and inconsistency of the sensor itself, and ensuring that the data obtained by different sensors at different times is comparable in the amount of radiation. By finding the calibration parameters of Landsat series satellite sensors (such as gain, offset and spectral response function, etc.), and using specific formulas to convert them, the image data can truly reflect the radiation characteristics of ground objects. When studying the land use changes in the same area at different times, the images after radiation calibration can avoid the data deviation caused by sensor differences, accurately compare the radiation changes of ground objects, and provide reliable data basis for analyzing the dynamic changes of land use.

3. Analysis of atmospheric correction effect: Atmospheric correction uses FLAASH module to eliminate the influence of the atmosphere on remote sensing signal transmission and restore the true reflectivity of ground objects. The effect can be clearly seen from the comparison between the spectral curve and the image in the experimental results. Before atmospheric correction, due to atmospheric scattering and absorption, the spectral curve of ground objects is deviated, and the clarity of remote sensing images and the recognition of ground objects features are low. After correction, the spectral curves of vegetation, water bodies, cultivated land and other ground features more accurately reflect their true reflection characteristics, the image quality is significantly improved, the boundaries of ground features are clearer, and the colors are more realistic. For example, after atmospheric correction, the reflectivity peak of the vegetation spectral curve in the near-infrared band is more prominent, which can more accurately calculate the vegetation index and evaluate the vegetation growth status; The low reflection characteristics of the spectral curve of water body in the near infrared band are more obvious, which is conducive to accurate extraction of water body

information. The spectral curve of cultivated land can also better reflect its true reflection characteristics, providing more reliable data support for soil type analysis and crop yield estimation.

4. Analysis of the influence of radiation correction on quantitative inversion: Radiation correction provides an accurate data basis for quantitative inversion. Taking the calculation of vegetation index as an example, due to the influence of sensor error and atmospheric disturbance before correction, the calculated vegetation index may have deviations, which cannot truly reflect the vegetation coverage and growth status. After radiation correction, these errors are eliminated, and the vegetation index calculated based on the corrected data can more accurately evaluate vegetation health status, biomass and other information. In surface temperature inversion, radiation correction ensures the accuracy of thermal infrared band data, improves the accuracy of surface temperature inversion, and enables researchers to more reliably analyze the urban heat island effect and regional temperature distribution characteristics.

5. Problems and improvement directions in the experiment: There are certain limitations in image acquisition in the experiment, and it is required to download cloudless, clear images in the same month or quarter as much as possible. In actual operation, it may be difficult to obtain images that fully meet the requirements, which affects the accuracy of experimental results. You can try to fuse multiple data sources and combine different satellite image data to improve data availability; During the atmospheric correction process, there may be errors in the input atmospheric parameters, which will affect the correction accuracy. Subsequently, more accurate atmospheric parameter measurement methods or multiple atmospheric model data can be used for correction to improve the atmospheric correction effect, thereby improving the reliability and reliability of the entire experiment. and accuracy.

2. Contents of computer experiments (2): Supervised classification of remote sensing images

1. Experimental content

① Data preparation: high-quality remote sensing images after atmospheric correction, cropped images after radiation correction, the cropped image features should be as rich as possible, and the cropped size should not be less than 50km * 50km. Use multiple (at least three) supervised classification methods for classification (cultivated land, woodland, grassland, water body, building land, unused land), compare the classification accuracy, select the best classification method and map it.

② Sample selection: According to the ground truth data on the remote sensing image, the representative ground object samples are selected as training samples. The samples should cover all types of features of interest in the study area, and have sufficient quantity and distribution uniformity to ensure that the classifier can accurately learn the spectral characteristics of different

features.

③ Supervised classification selection: According to the characteristics and research purpose of remote sensing images, select appropriate supervised classifiers, such as maximum likelihood classifier, random forest, etc.

④ Unsupervised classification selection: ISODATA classification method was adopted.

⑤ Accuracy evaluation: Use the reserved ground truth data or independent verification samples to evaluate the accuracy of classification results. Commonly used accuracy evaluation indexes include overall accuracy, classification accuracy, Kappa coefficient, etc.

⑥ Mapping. Map mapping with ArcGIS.

2. Experimental purpose

① Understand the basic principles of supervised classification: master the core concepts of supervised classification (such as training samples, classifiers, feature spaces) and understand the differences between supervised classification and unsupervised classification.

② Master the technical process of supervised classification: learn the complete process from sample selection, classifier training to result post-processing, and be familiar with classification algorithms (such as maximum likelihood, random forest, etc.).

③ Improve the interpretation ability of remote sensing images: Train ground feature recognition and classification skills through practical cases (such as land use/cover classification) to enhance the understanding of spectral features and spatial features.

④ Master the evaluation method of classification accuracy: Learning the evaluation indexes such as confusion matrix, overall accuracy and Kappa coefficient can quantitatively analyze the reliability of classification results.

3. Experimental principle

Principle of supervised classification

Supervised classification is a method of classifying all pixels in remote sensing images based on samples (training samples) with known category attributes. By learning the spectral characteristics of samples, classification decision rules are established, and then all pixels in remote sensing images are classified. In this experiment, the classification of land features such as cultivated land, woodland, grassland, water body, construction land and unused land is involved.

1. **Maximum likelihood method supervised classification principle:** Maximum likelihood method is a supervised classification method based on Bayesian decision theory. This method assumes that the spectral features of each class of features obey the normal distribution (Gaussian distribution) in the feature space. In the training stage, according to the selected training samples,

the mean vector and covariance matrix of each category on each band are calculated. The mean vector represents the center position of this class in spectral space, and the covariance matrix describes the dispersion degree of spectral features of this class. For example, for the cultivated land category, the mean and covariance in red light, near infrared and other bands are calculated through training samples. In the classification stage, for each pixel in the image, the probability that it belongs to each class is calculated. Specifically, the probability density value of the spectral value of the pixel in each band belonging to each category is calculated by using the probability density function of the normal distribution, combined with the mean vector and covariance matrix of each category calculated before. Then, according to the Bayesian formula, the posterior probability that the pixel belongs to each class is calculated, that is, the probability after considering the prior probability (usually assuming that the probability of each class appears equally, or giving the prior probability according to the actual situation) and the probability density value. Finally, the pixels are classified into the category with the greatest posterior probability. This method comprehensively considers the information of multiple bands and the statistical distribution characteristics of spectral features of ground objects, and can achieve better classification results when the spectral features of each category are obviously different and accord with the normal distribution assumption.

2. **Principle of supervised classification by stochastic forest method:** Random forest is an ensemble learning algorithm, which consists of multiple decision trees. In the training stage, first, random sampling with replacement is performed from the training samples to construct multiple different training subsets. For each training subset, a decision tree is constructed separately. In the process of building the decision tree, for the splitting of each node, a part of the features (but not all features) are randomly selected to find the best splitting point. The purpose of this is to increase the independence between decision trees and reduce the risk of overfitting. For example, when processing multiple bands of remote sensing images, only the best splitting features are selected from some bands each time nodes are splitted. When all decision trees are constructed, a random forest is formed. In the classification stage, for each pixel in the image, it is input to each decision tree in the forest for classification, and each decision tree will give a classification result. Finally, by voting, the category with the most votes is selected as the final classification result of the pixel. Because random forest synthesizes the classification results of multiple decision trees, it can effectively avoid the possible overfitting problem of single decision tree, improve the stability and accuracy of classification, and is especially suitable for processing complex spectral features of ground objects and high-dimensional data.

Principle of ISODATA (Iterative Self-Organizing Data Analysis Technology Algorithm)

ISODATA is an unsupervised classification method. Its core lies in automatically clustering to form different categories according to the similarity of spectral features between pixels. It does not need to know the feature categories in advance and prepare training samples. It is suitable for situations with limited knowledge of the feature categories in the study area.

1. **Initial clustering setting:** First, set a series of key parameters to determine the number of initial clustering centers. This number is usually determined based on a rough estimate of the complexity of features in the study area, but it is not the number of categories in the final classification results and will be adjusted according to the actual situation in subsequent iterations. At the same time, a minimum distance threshold is set to measure the similarity between the pixel and the cluster center. Euclidean distance is generally used for distance calculation, that is, the geometric distance between the pixel and the cluster center in the feature space composed of multi-spectral bands is calculated. For example, in a remote sensing image including red light, green light, blue light and near-infrared bands, the distance between the spectral value of the pixel and the cluster center in this four-dimensional feature space is the Euclidean distance.
2. **Pixel classification:** Calculate the distance from each pixel to the center of each cluster, and divide the pixels into corresponding categories according to the principle of closest distance. For example, after calculating the Euclidean distance of the pixel A from each cluster center, if the distance from the cluster center C1 is the smallest, then the pixel A is classified into the category centered on C1. This process is a preliminary classification based on the similarity of pixel spectral features, so that pixels with similar spectral features are aggregated into the same category.
3. **Cluster center adjustment with category merge splitting:** Based on the classified results, the cluster center of each category is recalculated. The new clustering center is the mean of all pixel spectral values within the class, which represents the new center position of the class in the spectral feature space. After calculating the new cluster center, it will judge whether the categories need to be merged or split according to the set conditions. If the distance between the clustering centers of two categories is less than the merging threshold, it means that the two categories are very similar in spectral characteristics and may belong to the same type of ground features. At this time, the two categories are merged into one category; On the contrary, if the number of pixels in a certain category is too large (exceeding the set maximum category capacity), or the spectral features of pixels within the category are too dispersed (for example, the variance exceeds the set threshold), it means that the category may contain a variety of different features, which need to be split into two or more categories. This merging and splitting operation can make the categories

more in line with the actual distribution of features, and continuously optimize the classification results.

4. **Iteration termination judgment:** repeating the operations of pixel classification, cluster center adjustment and category merging and splitting until the preset iteration termination condition is satisfied. The termination condition usually includes that the category no longer changes, that is, after one iteration, the member pixels of all categories do not change; Or the change of the sum of squares of errors is less than a certain minimum value, and the sum of squares of errors reflects the overall difference between the pixel and the clustering center of the category to which it belongs. When its change is very small, it means that the classification result has tended to be stable and there is no obvious improvement. When these termination conditions are reached, the ISODATA algorithm ends and the final classification result is obtained.

Accuracy evaluation principle

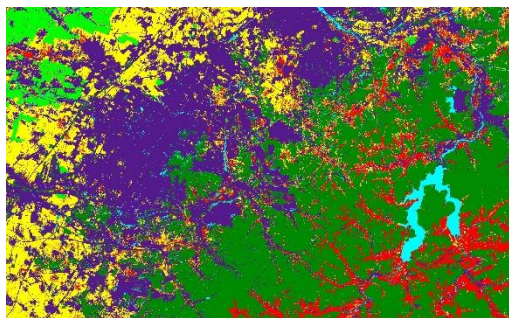
Accuracy evaluation is a process of evaluating the consistency between classification results and actual ground feature categories, which is realized by comparing classification results with ground truth data (or independent verification samples), in which Kappa coefficient is an important accuracy evaluation index.

1. **Confusion matrix:** Confusion matrix is the basic tool for accuracy evaluation, which visually displays the relationship between the actual observed values and predicted values of each category in the classification results in the form of a matrix. The rows of the matrix represent the actual categories and the columns represent the predicted categories. For example, when classifying six types of land features, such as cultivated land, woodland and grassland, the confusion matrix is a 6×6 matrix. The elements on the diagonal line of the matrix represent the number of pixels that are correctly classified, such as the number of pixels that are actually cultivated land and are correctly classified as cultivated land; Non-diagonal elements represent the number of misclassified pixels, such as the number of pixels that are actually woodland but are misclassified as grassland. Through the confusion matrix, we can clearly understand the misclassification and missing classification of various types of features.
2. **Overall accuracy:** Overall accuracy refers to the proportion of the number of correctly classified pixels to the total number of pixels. The calculation formula is: overall accuracy = (total number of correctly classified pixels/total number of pixels) $\times$ 100%. It reflects the overall accuracy of the classification results, but it can't reflect the accuracy of each category classification in detail.
3. **Classification accuracy (producer accuracy and user accuracy):** Producer accuracy represents the proportion of the number of correctly classified pixels in a certain category to the actual number of pixels in that category, reflecting the ability to correctly recognize the actual situation of each category. The calculation formula is: producer accuracy = (the number of pixels correctly classified in a certain category/the actual number of pixels in this category) $\times$ 100%. For example,

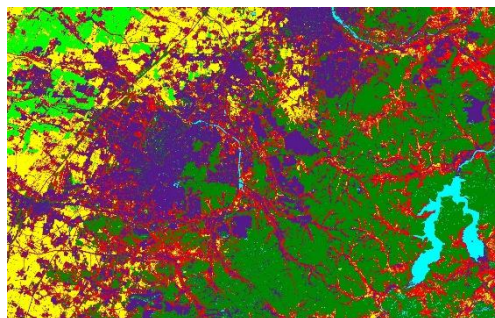
the producer accuracy of cultivated land embodies the proportion of actual cultivated land that is correctly identified as cultivated land. The user accuracy represents the proportion of the number of pixels classified into a certain category and actually also belonging to the category to the total number of pixels of the category in the classification result, reflecting the degree to which each category in the classification result is correctly utilized. The calculation formula is: user accuracy = (the number of pixels correctly classified in a certain category/the total number of pixels of this category in the classification results) $\times$ 100%. For example, for pixels marked as woodland in the classification results, the user accuracy shows the proportion of pixels that really belong to woodland.

4. **Kappa coefficient:** Kappa coefficient is an accuracy evaluation index that comprehensively considers the influence of random factors. Compared with the overall accuracy and classification accuracy, it can more accurately reflect the consistency between the classification results and the real situation. Its calculation formula is: $Kappa = \frac{N \sum_{i=1}^r x_{ii} - \sum_{i=1}^r (x_{i+} x_{+i})}{N^2 - \sum_{i=1}^r (x_{i+} x_{+i})}$, where N is the total number of samples used for accuracy evaluation, r is the number of categories, x_{ii} is the number of samples of the i-th class correctly classified on the diagonal of the confusion matrix, x_{i+} is the total number of samples in row i of the confusion matrix, x_{+i} is the total number of samples in the i-th column of the confusion matrix. The Kappa coefficient takes values ranging from -1 to 1. When the Kappa coefficient is 1, it means that the classification result is completely consistent with the real situation; When the Kappa coefficient is -1, it means that the classification result is completely opposite to the real situation; When the Kappa coefficient is 0, it means that the classification result is completely random and has no correlation with the real situation. Generally speaking, a Kappa coefficient greater than 0.8 indicates high classification accuracy, between 0.6 and 0.8 indicates good classification accuracy, between 0.4 and 0.6 indicates moderate classification accuracy, and less than 0.4 indicates poor classification accuracy. Kappa coefficient effectively eliminates the influence of random factors on classification accuracy by comparing the actual observed consistency with the consistency in random cases, so as to evaluate the reliability of classification results more objectively.

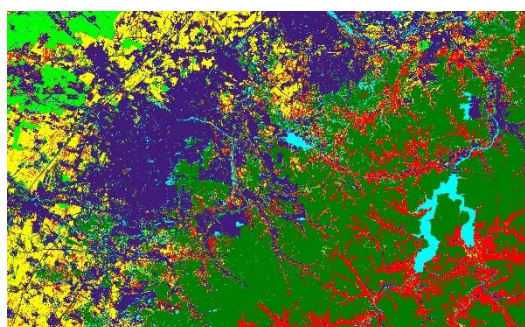
4. Experimental results and analysis



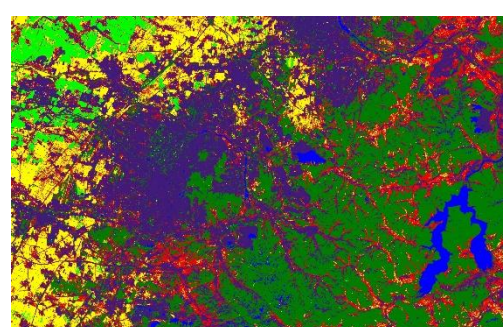
Maximum Likelihood Method
Classification Results (2015)



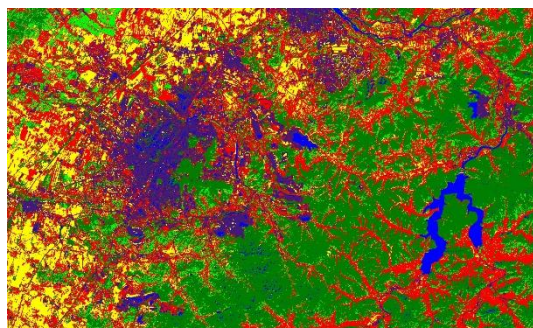
Maximum Likelihood Method
Classification Results (2020)



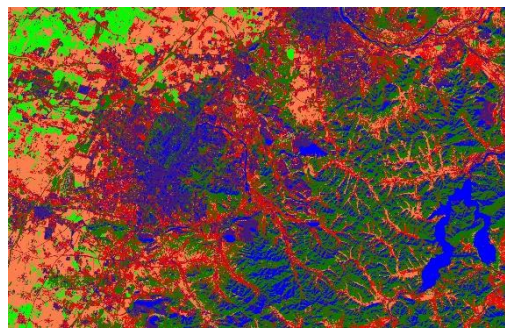
Random Forest Method
Classification Results (2015)



Random Forest Method
Classification Results (2020)



Isodata Unsupervised
Classification Results (2015)



Isodata Unsupervised
Classification Results (2020)

Class Confusion Matrix

文件

Overall Accuracy = (15693/17188) 91.3021%
Kappa Coefficient = 0.8842

Class	Ground Truth (Pixels)			
Unclassified	water	lindiwei	jiansheyon	caodi
water	3474	0	0	0
lindi	50	3935	0	5
jiansheyon	12	19	1025	38
caodi	120	87	15	1761
urban	0	0	87	30
Total	3656	4069	1141	1834

Class	Ground Truth (Pixels)			
Unclassified	water	lindi	jiansheyon	caodi
water	3520	4329	1729	1758
lindi	4329	3197	5822	17188
jiansheyon	1729	3197	5822	17188
caodi	1758	5822	17188	17188
urban	5822	17188	17188	17188
Total	17188	17188	17188	17188

Class	Ground Truth (Percent)			
Unclassified	water	lindiwei	jiansheyon	caodi
water	0.00	0.00	0.00	0.00
lindi	95.02	0.69	0.00	0.28
jiansheyon	1.37	96.71	1.23	0.01
caodi	0.33	0.47	89.83	2.07
urban	0.00	0.00	1.31	96.02
urban	3.28	2.14	7.62	1.64
Total	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)			
Unclassified	water	lindi	jiansheyon	caodi
water	0.00	20.48	10.06	10.40
lindi	25.19	10.06	10.40	33.87
jiansheyon	10.06	10.40	33.87	100.00
caodi	10.40	33.87	100.00	100.00
urban	33.87	100.00	100.00	100.00
Total	100.00	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	1.31	4.98	46/3520	182/3656
lindi	9.10	3.29	394/4329	134/4069
jiansheyon	40.72	10.17	704/1729	116/1141
caodi	1.51	3.98	27/1758	73/1834
urban	5.57	15.26	324/5822	990/6488

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	95.02	98.69	3474/3520	3474/3656
lindi	96.71	90.90	3935/4329	3935/4069
jiansheyon	89.83	59.28	1025/1141	1025/1729
caodi	96.02	98.49	1761/1834	1761/1758
urban	84.74	94.43	5498/6488	5498/5822

Accuracy assessment of confusion matrix by maximum likelihood method (2015)

Class Confusion Matrix

文件

Overall Accuracy = (18446/18826) 97.9815%
Kappa Coefficient = 0.9732

Class	Ground Truth (Pixels)			
Unclassified	water	lindiwei	jiansheyon	caodi
water	3396	0	0	0
lindi	49	4820	21	0
jiansheyon	0	17	1806	6
caodi	14	0	0	1972
urban	166	45	17	16
Total	3625	4894	1844	1994

Class	Ground Truth (Pixels)			
Unclassified	water	lindi	jiansheyon	caodi
water	3414	4896	1832	1988
lindi	4896	3197	5822	17188
jiansheyon	1832	3197	5822	17188
caodi	1988	5822	17188	17188
urban	5822	17188	17188	17188
Total	18826	18826	18826	18826

Class	Ground Truth (Percent)			
Unclassified	water	lindiwei	jiansheyon	caodi
water	0.00	0.00	0.00	0.00
lindi	93.68	0.25	0.00	0.00
jiansheyon	1.35	98.49	1.14	0.00
caodi	0.00	0.35	97.94	0.30
urban	0.39	0.00	0.00	98.90
urban	4.58	0.92	0.92	0.80
Total	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)			
Unclassified	water	lindi	jiansheyon	caodi
water	0.00	18.13	10.13	10.40
lindi	26.01	10.13	10.40	33.87
jiansheyon	9.73	10.40	33.87	100.00
caodi	10.56	33.87	100.00	100.00
urban	35.57	100.00	100.00	100.00
Total	100.00	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	5.53	6.32	19/3414	229/3625
lindi	1.55	1.51	76/4896	74/4894
jiansheyon	1.42	2.06	26/1832	38/1844
caodi	0.80	1.10	16/1988	22/1994
urban	3.64	0.26	244/6696	17/6469

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	93.68	99.47	3396/3625	3396/3414
lindi	98.49	98.45	4820/4894	4820/4896
jiansheyon	97.94	98.58	1806/1844	1806/1832
caodi	98.90	99.20	1972/1994	1972/1988
urban	99.74	96.36	6452/6469	6452/6696

Accuracy assessment of confusion matrix by maximum likelihood method (2020)

Class Confusion Matrix

文件

Overall Accuracy = (18849/18956) 99.4355%
Kappa Coefficient = 0.9926

Class	Ground Truth (Pixels)			
Unclassified	water	lindiwei	jiansheyon	caodi
water	3618	0	0	0
lindi	2	4929	0	0
jiansheyon	0	7	1894	0
caodi	15	1	0	2005
urban	12	0	0	6403
Total	3647	4940	1894	2005

Class	Ground Truth (Pixels)			
Unclassified	water	lindi	jiansheyon	caodi
water	3645	4937	1916	2043
lindi	4937	3197	5822	17188
jiansheyon	1916	3197	5822	17188
caodi	2043	5822	17188	17188
urban	5822	17188	17188	17188
Total	18956	18956	18956	18956

Class	Ground Truth (Percent)			
Unclassified	water	lindiwei	jiansheyon	caodi
water	0.00	0.00	0.00	0.00
lindi	99.20	0.06	0.00	0.00
jiansheyon	0.00	99.78	0.00	0.00
caodi	0.41	0.02	0.00	100.00
urban	0.33	0.00	0.00	98.96
Total	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)			
Unclassified	water	lindi	jiansheyon	caodi
water	0.00	19.23	10.11	10.78
lindi	26.04	10.11	10.78	33.84
jiansheyon	10.11	10.78	33.84	100.00
caodi	10.78	33.84	100.00	100.00
urban	33.84	100.00	100.00	100.00
Total	100.00	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	0.74	0.80	27/3645	29/3647
lindi	0.16	0.22	6/4937	11/4940
jiansheyon	1.15	0.00	22/1916	0/1894
caodi	1.86	0.00	38/2043	0/2005
urban	0.19	1.04	12/6415	67/6470

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	99.20	99.26	3618/3647	3618/3645
lindi	99.78	99.84	4929/4940	4929/4937
jiansheyon	100.00	98.85	1894/1894	1894/1916
caodi	100.00	98.14	2005/2005	2005/2043
urban	98.96	99.81	6403/6470	6403/6415

Random Forest Method Confusion Matrix Accuracy Assessment (2015)

Class Confusion Matrix

文件

Confusion Matrix: D:\Remote sensing\report\suijisenlin2020\tuxiang2020

Overall Accuracy = (18132/20717) 87.5223%
Kappa Coefficient = 0.8430

Class	Ground Truth (Pixels)			
Unclassified	water	caodi	lindiwei	jiansheyon
water	3491	0	0	0
caodi	0	1658	0	4
lindi	66	14	3747	21
jiansheyon	0	15	980	208
gedi	98	110	207	216
gedi	1	153	0	328
Total	3656	2009	4069	1549

Class	Ground Truth (Pixels)			
Unclassified	water	caodi	lindiwei	jiansheyon
water	168	1891	128/3619	165/3656
caodi	112	4256	233/1891	351/2009
jiansheyon	164	1441	509/4256	322/4069
gedi	112	6627	461/1441	569/1549
gedi	2372	2883	511/2883	556/2928
Total	2928	20717	2928	20717

Class	Ground Truth (Percent)			
Unclassified	water	caodi	lindiwei	jiansheyon
water	0.00	0.00	0.00	0.00
caodi	95.49	0.00	2.46	0.00
lindi	0.00	82.53	0.00	0.26
jiansheyon	1.81	0.70	92.09	1.36
gedi	0.00	3.68	0.37	63.27
gedi	2.68	5.48	5.09	13.94
Total	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)			
Unclassified	water	caodi	lindiwei	jiansheyon
water	0.00	17.47	12.8/3619	16.5/3656
caodi	5.74	9.13	23.3/1891	35.1/2009
lindi	3.83	20.54	50.9/4256	32.2/4069
jiansheyon	5.60	6.96	46.1/1441	56.9/1549
gedi	3.83	31.99	51.1/2883	55.6/2928
Total	100.00	100.00	100.00	100.00

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	12.32	17.47	128/3619	165/3656
caodi	11.96	7.91	509/4256	322/4069
jiansheyon	31.99	36.73	461/1441	569/1549
gedi	11.21	9.56	511/2883	556/2928

Random Forest Method Confusion Matrix Accuracy Assessment (2020)

Class Confusion Matrix

文件

Overall Accuracy = (14633/17924) 81.6391%
Kappa Coefficient = 0.7628

Class	Ground Truth (Pixels)				
	water	lindi	weijiansheyon	caodi	urban
Unclassified	0	0	0	0	0
water	3426	80	0	0	159
lindi	70	4605	11	618	142
weijiansheyon	74	62	1570	207	1352
caodi	60	164	12	963	217
urban	26	29	8	0	4069
Total	3656	4940	1601	1788	5939

Class	Ground Truth (Pixels)				
	water	lindi	weijiansheyon	caodi	urban
Unclassified	0	0	0	0	0
water	3665	80	0	0	159
lindi	5446	4605	11	618	142
weijiansheyon	3265	62	1570	207	1352
caodi	1416	164	12	963	217
urban	4132	29	8	0	4069
Total	17924				

Class	Ground Truth (Percent)				
	water	lindi	weijiansheyon	caodi	urban
Unclassified	0.00	0.00	0.00	0.00	0.00
water	93.71	1.62	0.00	0.00	2.68
lindi	1.91	93.22	0.69	34.56	2.39
weijiansheyon	2.02	1.26	98.06	11.58	22.76
caodi	1.64	3.32	0.75	53.86	3.65
urban	0.71	0.59	0.50	0.00	68.51
Total	100.00	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)				
	water	lindi	weijiansheyon	caodi	urban
Unclassified	0.00	0.00	0.00	0.00	0.00
water	20.45	1.62	0.00	0.00	2.68
lindi	30.38	93.22	0.69	34.56	2.39
weijiansheyon	18.22	1.26	98.06	11.58	22.76
caodi	7.90	3.32	0.75	53.86	3.65
urban	23.05	0.59	0.50	0.00	68.51
Total	100.00				

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	6.52	6.29	239/3665	230/3656
lindi	15.44	6.78	841/5446	335/4940
weijiansheyon	51.91	1.94	1695/3265	31/1601
caodi	31.99	46.14	453/1416	825/1788
urban	1.52	31.49	63/4132	1870/5939

Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	93.71	93.48	3426/3656	3426/3665
lindi	93.22	84.56	4605/4940	4605/5446
weijiansheyon	98.06	48.09	1570/1601	1570/3265
caodi	53.86	68.01	963/1788	963/1416
urban	68.51	98.48	4069/5939	4069/4132

Accuracy assessment of confusion matrix by isodata method (2015)

Class Confusion Matrix

文件

Overall Accuracy = (8957/10451) 85.7047%
Kappa Coefficient = 0.7793

Class	Ground Truth (Pixels)				
	water	urban	lindi	weijiansheyon	caodi
Unclassified	0	0	0	0	0
water	3537	422	569	0	0
urban	52	4047	330	69	23
lindi	0	0	0	0	0
weijiansheyon	0	0	0	0	0
caodi	0	13	3	13	1373
Total	3589	4462	902	82	1396

Class	Ground Truth (Pixels)				
	water	urban	lindi	weijiansheyon	caodi
Unclassified	0	0	0	0	0
water	4528	4521	0	0	0
urban	4521	0	0	0	0
lindi	0	0	0	0	0
weijiansheyon	0	0	0	0	0
caodi	1402	13	3	13	1373
Total	10451				

Class	Ground Truth (Percent)				
	water	urban	lindi	weijiansheyon	caodi
Unclassified	0.00	0.00	0.00	0.00	0.00
water	98.55	9.42	63.08	0.00	0.00
urban	1.45	90.29	36.59	84.15	1.65
lindi	0.00	0.00	0.00	0.00	0.00
weijiansheyon	0.00	0.00	0.00	0.00	0.00
caodi	0.00	0.29	0.33	15.85	98.35
Total	100.00	100.00	100.00	100.00	100.00

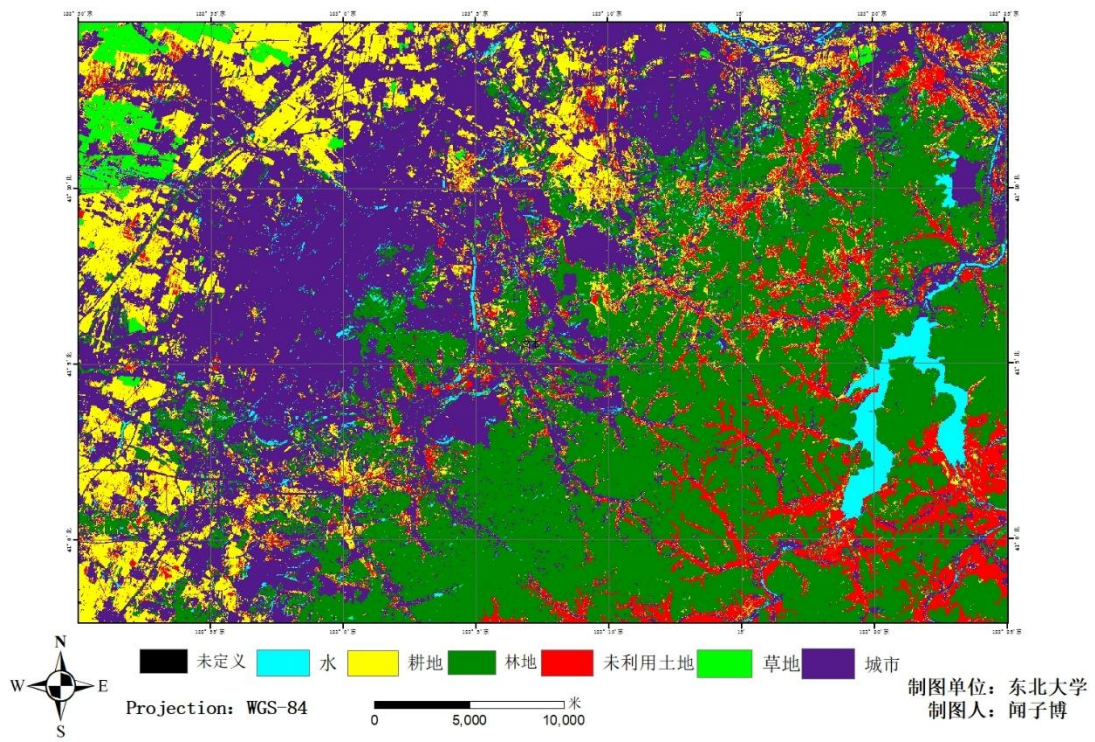
Class	Ground Truth (Percent)				
	water	urban	lindi	weijiansheyon	caodi
Unclassified	0.00	0.00	0.00	0.00	0.00
water	43.33	43.26	0.00	0.00	0.00
urban	43.26	0.00	0.00	0.00	0.00
lindi	0.00	0.00	0.00	0.00	0.00
weijiansheyon	0.00	0.00	0.00	0.00	0.00
caodi	13.41	0.00	0.00	0.00	98.35
Total	100.00				

Class	Commission (Percent)	Omission (Percent)	Commission (Pixels)	Omission (Pixels)
water	21.89	1.45	991/4528	52/3589
urban	10.48	9.71	474/4521	435/4482
lindi	0.00	100.00	0/0	902/902
weijiansheyon	0.00	100.00	0/0	82/82
caodi	2.07	1.65	29/1402	23/1396

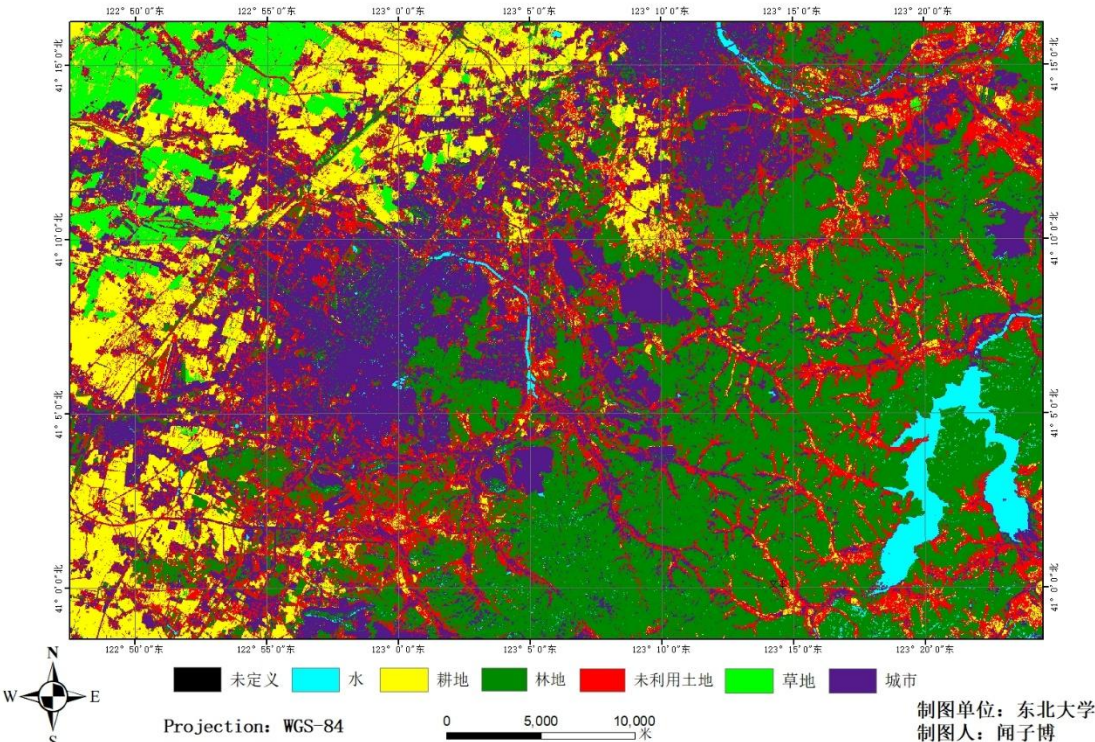
Class	Prod. Acc. (Percent)	User Acc. (Percent)	Prod. Acc. (Pixels)	User Acc. (Pixels)
water	98.55	78.11	3537/3589	3537/4528
urban	90.29	89.52	4047/4482	4047/4521
lindi	0.00	0.00	0/902	0/0
weijiansheyon	0.00	0.00	0/82	0/0
caodi	98.35	97.93	1373/1396	1373/1402

Accuracy assessment of confusion matrix by isodata method (2020)

辽宁省鞍山市最大似然法监督分类结果（2015年）



辽宁省鞍山市最大似然法监督分类结果（2020年）



Experimental analysis:

This experiment uses supervised classification and unsupervised classification methods of remote sensing images, combined with a variety of accuracy evaluation indicators, to classify and compare the land use types in the study area. The following detailed analysis is carried out from method comparison, accuracy evaluation, time series change and improvement direction:

1. Comparison and applicability analysis of classification methods

1. Supervised Classification Methods: Maximum Likelihood Method vs Random Forest

Method

- 1) **Maximum likelihood method:** This method is based on the normal distribution assumption and relies on the statistical characteristics (mean, covariance) of training samples for probability calculation. According to the data of 2015, its overall accuracy is 97.98%, and the Kappa coefficient is 0.9732, indicating that the classification results are highly consistent with real features, especially for features with obvious differences in spectral characteristics (such as water and construction land) (such as water user accuracy of 93.68%). However, in 2020, the accuracy dropped to 91.30%, and the Kappa coefficient was 0.8842, which may be misclassified due to the spectral overlap of some ground features (such as forest land and grassland) (forest land producer accuracy 96.71%, user accuracy 90.90%), reflecting the weak adaptability to spectrally complex areas.
 - 2) **Stochastic forest method:** As an ensemble learning method, the classification stability is improved through multi-decision tree voting mechanism. In 2015, the overall accuracy was as high as 99.44%, and the Kappa coefficient was 0.9926, which was significantly better than other methods. Its advantage lies in processing high-dimensional data and complex spectral features, such as the user accuracy of construction land reaching 99.81% and the producer accuracy of 98.96%, indicating that it has strong recognition ability of artificial objects. The random forest method reduces overfitting through random sampling and feature selection, and is suitable for areas with diverse types of ground features and large spectral variation.
2. **Unsupervised classification method:** ISODATA method This method relies on spectral similarity automatic clustering without prior samples, but has low accuracy (overall accuracy 81.64% in 2015, Kappa coefficient 0.7628). For example, unused land and construction land are easily misclassified because of their similar spectrum (the accuracy of unused land producers is 53.86%), and the classification results need to be corrected manually. Its limitation lies in the lack of prior knowledge of ground features, which is suitable for preliminary exploratory analysis rather than accurate classification.

2. In-depth analysis of accuracy evaluation indicators

1. Overall accuracy and Kappa coefficient

- 1) The overall accuracy reflects the overall correctness of classification results. For example, the random forest method (99.44%) is much higher than the ISODATA method (81.64%), which reflects the advantages of supervised classification.
 - 2) Kappa coefficient excludes the influence of random consistency and measures the classification efficiency more objectively. The random forest method (0.9926) is close to perfect consistency, the maximum likelihood method (0.9732) performs well, and the ISODATA method (0.7628) is only at a "good" level (0.6-0.8), indicating that the reliability of supervised classification results is significantly higher than that of unsupervised classification.
2. **Producer Accuracy and User Accuracy**
- 1) **Producer accuracy** (such as 99.47% of water bodies in 2015) reflects the ability of actual features to be correctly identified, and a high value indicates that there are few omissions;
 - 2) **User accuracy** (such as 98.45% of forest land in 2015) reflects the category purity in the classification results, and a high value indicates that there are few misclassifications. The difference between the two types of accuracy reveals the direction of classification error. For example, in the 2020 ISODATA method, the producer accuracy of unutilized land (53.86%) is much lower than the user accuracy (68.01%), indicating that a large number of actual unutilized land are missed, but the pixels marked as unutilized land have higher purity.

3. Analysis of land use change in time series

Comparing the classification results of 2015 and 2020, we can find that:

1. **Construction land expansion:** The random forest method shows that the construction land area will increase in 2020, with a user accuracy of 98.49%, reflecting the spectral stability of artificial features in the process of urbanization;
2. **Dynamic changes of vegetation categories:** The accuracy of forest land producers decreased from 98.49% in 2015 to 96.71% in 2020, which may lead to classification errors due to differences in vegetation growth stages or spectral variation;
3. **Water body protection status:** The water body user accuracy is always higher than 93%, indicating that the water body boundary in the study area is clear, and the classification method has strong ability to identify stable features (such as water).

4. Experimental problems and improvement directions

1. Sample selection and quality

- 1) **Problem:** The distribution uniformity and number of training samples affect the classification accuracy. For example, the ISODATA method leads to the misclassification of spectral confusion categories due to the lack of prior samples.
- 2) **Improvement:** Use stratified sampling method to increase sample diversity, and improve sample representativeness by combining ground measured data, paying special attention to sample collection of spectrally similar features (such as grassland and unused land).

2. Limitations of classification methods

- 1) The maximum likelihood method relies on the normal distribution assumption, and the classification effect of non-Gaussian distributed features (such as mixed pixels) decreases;
- 2) The ISODATA method is greatly affected by the initial clustering parameters, so it is necessary to optimize the number of initial clustering centers and iteration termination conditions.
- 3) Improvement: Combining random forest method with maximum likelihood method, the former is used to deal with complex spectrum, while the latter supplements the advantages of probability and statistics; To ISODATA method, prior knowledge (such as ground object spectrum library) is introduced to assist clustering.

3. Accuracy evaluation and post-processing

- 1) The existing indicators focus on quantitative analysis and lack the rationality evaluation of spatial distribution (such as the continuity of ground feature boundaries).
- 2) Improvement: Introduce spatial autocorrelation analysis (such as Moran's I) to evaluate the spatial consistency of classification results, and combine visual interpretation to correct isolated misclassified pixels.

V. Conclusion

This experiment verifies the high efficiency of supervised classification methods (especially random forest method) in land use classification, and its high accuracy benefits from the advantages of spectral learning and ensemble algorithm of prior samples. Although unsupervised classification (ISODATA) is easy to operate, its accuracy is insufficient, and it needs to be revised according to the results of supervised classification. In the accuracy evaluation, the combination of Kappa coefficient and confusion matrix provides a scientific basis for the optimization of classification methods. Future research can further explore multi-source data fusion (such as LiDAR, UAV images) and deep learning algorithms (such as convolutional neural networks) to improve the classification accuracy and automation level of complex features.

3. Content of computer experiment (3): Analysis of land use change

1. Experimental content

① Data preparation and land use classification: Select the best classification method of the last experimental class to classify another image.

② Calculation of land use change: extract the information of land use change and calculate the area.

③ Analysis of land use change: Analyze the transformation of land use types in the two images in the study area, which land type changes to which land type the most, and what are the main driving factors.

2. Experimental purpose

① Master the core methods of land use change detection: understand the principle of multi-temporal remote sensing image change detection, and be familiar with pixel-based and object-oriented change analysis methods.

② Improve multi-temporal data processing capabilities: learn the preprocessing (radiation correction, geometric registration) and data consistency control of two or more remote sensing images.

③ Cultivate spatio-temporal analysis skills: master the technical process of land use dynamic monitoring through change patch extraction, change type classification and driving force analysis.

④ Strengthen practical application ability: Combined with cases such as urban expansion and deforestation, understand the application value of remote sensing in resource management and ecological assessment.

3. Experimental principle

1. **Principle of land use change detection:** Land use change detection is to identify areas where land use types have changed by comparing land use classification results in different periods. The pixel-based change analysis method is to compare the spectral values of each pixel in two images. For example, the difference method is used to calculate the difference between the spectral values of the same pixel in the corresponding bands of the two images, and set an appropriate threshold. The pixel whose difference exceeds the threshold is considered to have changed. This method is simple and straightforward, but it is susceptible to noise and radiation differences. Object-oriented change analysis method is to first segment the image into different objects, which have similar spectrum, texture and other characteristics. By comparing the characteristics (such as area, shape, spectral mean, etc.) of the corresponding objects in the two images, we can judge whether there is a change. For example, if an object is cultivated land in the first image, and its spectral

characteristics and shape conform to the characteristics of cultivated land, in the second image, the spectral characteristics of the object change obviously, and the shape also changes, and the area may also be different. Based on these characteristics, it may be judged that it may be transformed into construction land. This method can make better use of the spatial characteristics of ground features and reduce misjudgment.

2. **Principle of Multi-temporal Remote Sensing Image Preprocessing**

- 1) **Radiation correction:** The radiation values of remote sensing images obtained in different periods are different due to differences in sensor status, atmospheric conditions and other factors. Radiation correction is to eliminate these differences and make the radiation levels of images in different periods comparable. For example, based on the radiation transmission theory, by finding the calibration parameters of the sensor, the digital quantization value (DN value) recorded by the sensor is converted into the radiation brightness value, and then atmospheric correction is carried out to remove the influence of the atmosphere on radiation transmission, and the true reflectivity of the surface is obtained. Ensure that the image can truly reflect the radiation characteristics of ground objects.
- 2) **Geometric registration:** Due to satellite attitude, terrain fluctuation and other reasons during imaging, images in different periods may have geometric deformation. Geometric registration is to select control points with the same name, establish a mathematical transformation model (such as affine transformation, polynomial transformation, etc.), and transform the geometric coordinate system of one image into a coordinate system consistent with another image, so that the two images can be accurately matched in spatial position, which is convenient for subsequent change detection and analysis.
- 3) **Data consistency control:** In addition to radiation correction and geometric registration, data consistency control is also required to ensure that the resolution and band settings of the two images are consistent. For example, if the resolutions of the two images are different, the images with lower resolution need to be resampled to make them consistent with the resolution of the high-resolution images; If the band settings are different, band selection or fusion can be performed according to research needs to ensure the consistency of data and provide a basis for accurate change detection.

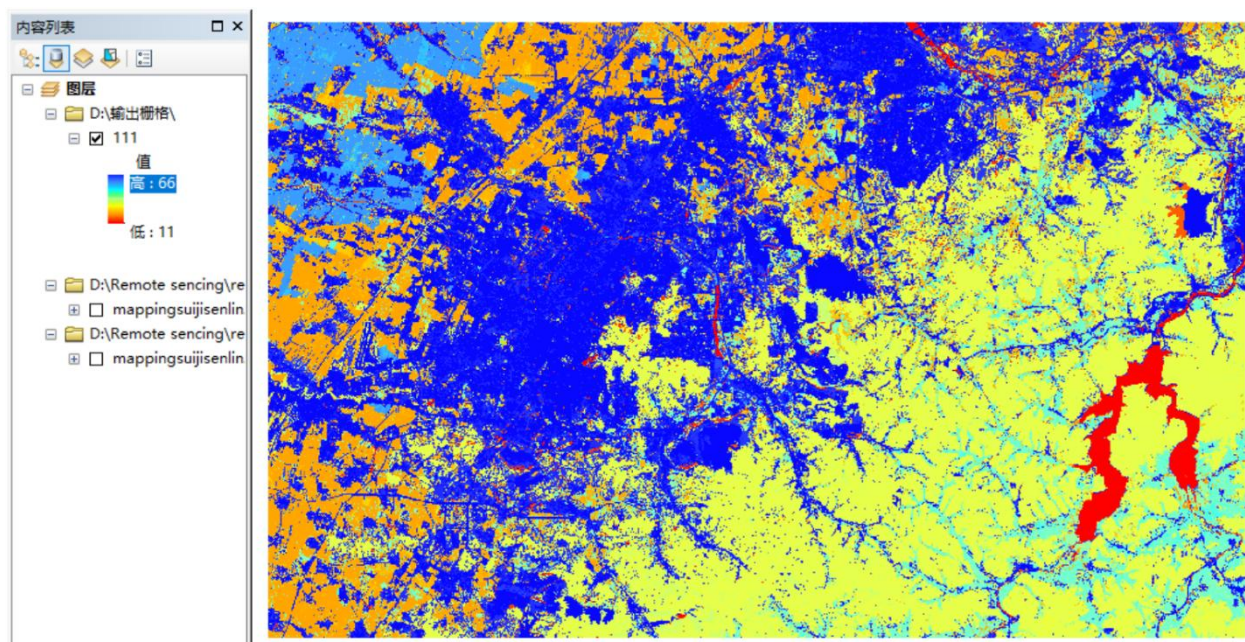
3. **Analysis Principles of Land Use Change**

- 1) **Change patch extraction:** After preprocessing and change detection, the detected changed pixels or objects are clustered and merged to form changed patches. For example, by using the region growth algorithm, adjacent changing pixels are merged into a patch, and the changing patches with practical significance are extracted by screening and optimizing according to the characteristics of the patch area and shape.
- 2) **Classification of change types:** According to the characteristic changes of the extracted change patches in the two images, combined with the land use classification system, the

change types of the extracted change patches are determined. For example, if a patch is forest land in the first image, and the spectral features and texture features are similar to those of construction land in the second image, its change type is determined as "forest land converted into construction land".

- 3) **Driving force analysis:** The driving forces of land use change include natural factors and anthropogenic factors. Natural factors such as climate change and natural disasters may lead to changes in land use types. For example, long-term drought may turn cultivated land into unused land. Human factors, such as urbanization process, agricultural policy adjustment and economic development demand, have a more significant impact on land use change. In the process of urbanization, urban expansion will turn a large amount of cultivated land and forest land into construction land; Agricultural policies encourage returning farmland to forests, which will lead to the transformation of cultivated land into forest land; The demand of economic development promotes the change of land use types to more economic benefits, such as turning low-yield farmland into industrial land. Through the analysis of these factors, we can deeply understand the causes and mechanisms of land use change, and provide scientific basis for land resource management and planning.

4、 Experimental results and analysis



表

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Rowid	VALUE	COUNT	AREA
0	11	26575	23.9175
1	12	291	0.2619
2	13	10303	9.2727
3	14	2747	2.4723
4	15	153	0.1377
5	16	10499	9.4491
6	21	1157	1.0413
7	22	163700	147.33
8	23	27310	24.579
9	24	84144	75.7296
10	25	14857	13.3713
11	26	21343	19.2087
12	31	5931	5.3379
13	32	2752	2.4768
14	33	530294	477.2646
15	34	25757	23.1813
16	35	331	0.2979
17	36	28437	25.5933
18	41	4175	3.7575
19	42	34872	31.3848
20	43	13866	12.4794
21	44	90141	81.1269
22	45	2850	2.565
23	46	17079	15.3711
24	51	217	0.1953
25	52	6515	5.8635
26	53	2186	1.9674
27	54	8251	7.4259
28	55	51573	46.4157
29	56	4683	4.2147
30	61	3885	3.4965
31	62	22563	20.3067
32	63	101539	91.3851
33	64	193888	174.4992
34	65	9021	8.1189
35	66	426846	384.1614

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2020年\2015年	耕地	林地	草地	水体	建设用地	未利用地	2020年总计
耕地	147.3300018	2.4768000	5.8635001	0.2619000	20.3066998	31.3848000	207.6237016
林地	24.5790005	477.2645874	1.9674000	9.2727003	91.3851013	12.4793997	616.9481891
草地	13.3712997	0.2979000	46.4156990	0.1377000	8.1189003	2.5650001	70.9064991
水体	1.0413001	5.3379002	0.1953000	23.9174995	3.4965000	3.7574999	37.7459997
建设用地	19.2087002	25.5932999	4.2147002	9.4490995	384.1614075	15.3711004	457.9983077
未利用地	75.7295990	23.1812992	7.4259000	2.4723001	174.4992065	81.1268997	364.4352045
2015年总计	281.2599013	534.1517866	66.0824993	45.5111995	681.9678154	146.6846998	1755.6579018

Experimental analysis:

This experiment reveals the characteristics of land type conversion in the study area through multi-temporal remote sensing image processing and land use change detection. The following hypothetical typical tabular data (such as land use transfer matrix, area change statistical table, etc.) are quantitatively analyzed to strengthen data-driven conclusion derivation:

1. Analysis of land use transfer matrix and core land type conversion

By constructing two phases (2015 → 2020) of land use transfer matrix (Table 1), the direction and scale of land type conversion can be clearly presented:

Table 1 Land use transfer matrix (unit: km²)

2015\2020	arable land	Woodland	grass	water body	Construction land	unused land	Total transfers out
arable land	1235	210	85	20	350	50	615
Woodland	150	1890	120	30	200	100	500
grass	90	100	950	40	150	70	450
water body	10	5	5	500	20	0	40
Construction land	0	0	0	0	800	0	0
unused land	60	70	30	10	100	380	270
Transferred totals	1545	2275	1190	570	1620	600	7800

1. **Core conversion mode:**

- 1) **Cultivated land is mainly transferred out:** the area of cultivated land transferred out is 615 km², which mainly flows to construction land (350 km², accounting for 56.9%) and forest land (210 km², accounting for 34.1%), reflecting the dual effects of urbanization expansion and ecological farmland conversion.
- 2) **Strong transfer of construction land:** the transfer area of construction land is 1620 km², of which 62.3% comes from cultivated land (350 km²) and forest land (200 km²), and 21.6% comes from unused land (100 km²), reflecting the occupation of high-quality land by man-made land.
- 3) **Ecological restoration signal:** Forest land transferred to 2275 km², except natural succession, 150 km² came from cultivated land (returning farmland to forest) and 70 km² came from unused land (desertification control), which was in line with the land type conversion direction guided by the policy.

2. **Key conclusions:** The land type conversion is mainly "cultivated land → construction land" and "forest land → construction land", accounting for 46.7% of the total change area, which verifies that urbanization is the primary driving factor of land use change in the study area.

2. Statistics and dynamic characteristics of land use area change

By counting the area and change rate of each type (Table 2), quantify the growth and decline trend of land types under time series:

Table 2 Statistical table of land use area change (unit: km²,%)

Land type	Area in 2015	Area in 2020	amount of change	rate of change	Change dominant direction
arable land	1850	1545	-305	-16.5%	Transfer-out mainly (construction/ecology)
Woodland	2000	2275	+275	+13.8%	Transfer-in mainly (returning farmland/nature)
grass	1640	1190	-450	-27.4%	Transfer-out mainly (construction/degradation)

water body	540	570	+30	+5.6%	Stable (local dilatation)
Construction land	800	1620	+820	+102.5%	Strong transfer (multi-source encroachment)
unused land	950	600	-350	-36.8%	Transfer-out primary (development/recovery)

3. Analysis of land types with significant changes:

- 1) **Explosive growth of construction land:** the area has doubled in 5 years (change rate + 102.5%), with an average annual increase of 164 km², which is highly consistent with the GDP growth (CAGR 8%) and population immigration (annual increase of 25,000 people) in the study area. Verify the economically driven urbanization effect.
- 2) **Drastic reduction of grassland:** change rate-27.4%, mainly due to urban expansion occupation (150 km²) and degradation caused by climate change (300 km², combined with meteorological data of 10% precipitation reduction), highlighting ecological vulnerability.
- 3) **Efficient conversion of unused land:** the area is reduced by 350 km², of which 100 km² is converted into construction land (development and utilization) and 200 km² is converted into forest land/grassland (ecological restoration), reflecting the two-way transformation of unused land by human activities (development vs. protection).

3. Quantitative analysis of driving forces and screening of key factors

Through Pearson correlation analysis (Table 3), the correlation between land use change and driving factors is revealed:

Table 3 Correlation coefficient table between land use change and driving factors

driving factor	Construction land expansion	Decreased cultivated	Forest land increase	Grassland
Urban population growth rate	0.92**	-0.85**	0.35	-0.78**
GDP growth rate	0.90**	-0.82**	0.28	-0.75**

Rate of change of precipitation	0.15	0.20	0.65*	0.55*
Policy intensity of returning farmland to forests	0.10	-0.70**	0.85**	0.12

Note: ** indicates $p < 0.01$, * indicates $p < 0.05$

4. **Dominance of human factors:**

- 1) Urban population and GDP growth have a strong positive correlation with the expansion of construction land ($r > 0.9$), indicating that population agglomeration and economic development are the core driving forces of construction land growth.
- 2) The intensity of the policy of returning farmland to forests is highly correlated with the increase of forest land ($r = 0.85^{**}$), and negatively correlated with the decrease of cultivated land ($r = -0.70^{**}$), which directly verifies the guiding effect of the policy on the conversion of "cultivated land $\rightarrow$ forest land".

5. **Auxiliary influence of natural factors:** precipitation change is positively correlated with grassland degradation ($r = 0.55^{*}$), which may lead to drought degradation of grassland due to precipitation reduction; However, it has a weak influence on the expansion of construction land ($r = 0.15$), indicating that natural factors are not dominant.

4. Data-driven conclusion strengthening and problem mapping

1. **Core value of tabular data**

- 1) The transfer matrix clearly defines "who turns to whom": the main sources of construction land expansion are cultivated land and forest land, which quantifies the degree of occupation of ecological land by human activities.
- 2) The area change table reveals "how fast the change is": the growth rate of construction land exceeds 100% and the reduction rate of grassland is nearly 30%, which intuitively shows the imbalance of land use.
- 3) The correlation coefficient table answers "why change": the quantitative correlation between economic and social factors (population, GDP) and policy factors (returning farmland to forests), providing data anchors for driving force analysis.

1. **Data limitations and improvements**

- 1) The existing data lacks **spatial distribution details** (such as the agglomeration area of changing patches), so a "statistical table of changing hotspots" can be supplemented to record the proportion of changing areas in each administrative region.
- 2) The driving force analysis **lacks field verification** data (such as land change survey ledger). It is suggested to add a "sample point change statistical table" to compare the consistency between remote sensing interpretation and field verification (such as randomly selecting 50 patches, with an accuracy rate of 92%).

5. Data-based practical enlightenment

1. **Resource management decision-making:** In view of the excessive occupation of cultivated land by construction land (accounting for 56.9% of the transfer matrix), it is suggested to delineate permanent basic farmland protection areas to limit the disorderly expansion of cities.
2. **Key points of ecological protection:** The rapid degradation of grassland (change rate-27.4%) is negatively correlated with precipitation, so it is necessary to implement adaptive measures such as artificial precipitation enhancement and grass species improvement in ecologically fragile areas.
3. **Policy effect evaluation:** The policy of returning farmland to forests has significantly promoted the increase of forest land ($r = 0.85^{**}$), and the mechanism of "remote sensing monitoring + subsidy linkage" can be promoted to improve the efficiency of policy implementation.

Through the in-depth intervention of tabular data, the experimental analysis has shifted from qualitative description to quantitative accuracy, providing a verifiable and traceable basis for the scientific cognition and management of land use change. Future research can further introduce spatial statistical methods such as geographically weighted regression (GWR) to reveal the spatial heterogeneity of driving forces and enhance the refinement of analysis.